

Learning Module Implementation Handbook

Deterministic core with optional HF-managed advisory AI

ARCHITECTURE

HF-MANAGED

EVIDENCE-FIRST

PRODUCTION GATES

Authoritative decision. VeriTrust executes no learned production inference in browser code, Vercel functions, platform workers, local runtimes, or VeriTrust-owned model servers. Learned production inference uses Hugging Face-managed infrastructure. The canonical hierarchy is HF-operated hf-inference when the exact model/task and all gates pass; then Hugging Face Inference Endpoint; then private Hugging Face repository plus Hugging Face Inference Endpoint. Third-party Hugging Face Inference Providers are DISABLED_BY_DEFAULT, outside the authoritative production path, and never an automatic fallback.

Status: implementation architecture and research package; no production performance claim. All UNKNOWN values remain explicit.

Prepared from the executable repository, seven historical PDFs, fresh official Hugging Face research, primary competitor sources and C2PA standards.

Contents and control

This is an Implementation Architecture Handbook, not a standalone executable specification. Critical invariants, model identities, deployment classes, evidence semantics, receipt fields, lifecycle states and companion references are generated from validated canonical structured sources; all provider availability is point-in-time and promotion-gated.

NORMATIVE IMPLEMENTATION COMPANIONS: `Architecture_Source_of_Truth.yaml`; `Model_Registry_Proposal.json`; `HuggingFace_Deployment_Matrix.csv`; `Implementation_Backlog.csv`; `Database_Target_Contract.md`; `Verification_Test_Plans.md`; `Migration_Roadmap.md`.

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Cross-document invariants

- HF-managed learned inference only; no normal third-party provider tier.
- Registry resolution precedes dispatch.
- Receipt identifies exact deployment, bundle and ordered learned components.
- `raw_output`, `raw_label`, `raw_score`, `normalized_label`, `normalized_score`, `calibrated_probability` and `calibration_version` remain distinct.
- Expected-evidence lifecycle uses the canonical 16-state enum: `planned`, `dispatched`, `running`, `completed`, `abstained`, `not_applicable`, `unavailable`, `timed_out`, `failed`, `cancelled`, `privacy_blocked`, `policy_skipped`, `insufficient`, `unsupported`, `degraded`, `budget_exceeded`.
- Specialist evidence cannot silently substitute or own final policy.
- No original PDF or production source file is overwritten.

1. Executive decision

No learned model is required for core learning. Optional AI is advisory-only and HF-managed.

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The current learning API/repository is a deterministic Supabase-backed application. Courses, lessons, enrollment, progress, assessments, certificates, assignments, cohorts and administration must remain fully functional when HF is unavailable.

2. Repository-grounded current behavior

`api/learning.js` and `lib/learning/repository.js` implement the learning domain through tables and RPCs. Front-end assessment flows do not expose correct answers before submission. The database manifest is descriptive and migrations are external.

Area	Current capability	Target
Catalog/content	courses and lessons	immutable published versions
Activity	enrollment/progress	server-owned state machine
Assessment	attempt/submission	versioned prompt snapshot and scoring
Credential	certificate records	revocable verifiable contract
Admin	authoring and operations	RBAC, audit, tenant scope

3. Target modular monolith

Keep learning modules in the existing application/API and database with strict boundaries. Separate catalog/publication, enrollment/assignment, progress/labs, assessment, credential and administration contracts. Background work handles reminders, exports and issuance; it does not own the only copy of state.

LEARNER / ADMIN -> LEARNING API
|- catalog + immutable content versions
|- enrollments + assignments + cohorts
|- progress + labs
|- attempts + scoring
|- credentials
|- audit + analytics

4. Catalog, lessons and publication

Draft, review, approve, schedule, publish and archive are server-enforced. Published content is immutable; correction creates a new version. Prerequisites and estimates are validated. Accessibility metadata, transcript, captions, keyboard paths and alternatives are part of publication gates.

- No client-only prerequisite or completion authority.
- Content references are tenant-scoped and versioned.
- Archive preserves learner and audit history.

5. Enrollment, assignments and progress

Enrollment and assignment transitions are idempotent. Progress events are append-only observations; a materialized state is derived server-side. Cohort/program rules, due dates, grace, reminders and accommodations are explicit. Concurrency tests prevent duplicate completion or lost updates.

State family	States
Enrollment	eligible, enrolled, active, completed, withdrawn
Assignment	assigned, accepted, in_progress, submitted, overdue, waived
Lesson	not_started, in_progress, completed

6. Assessment and certification

No recommendation or generative model may alter a score, pass/fail, credential or access decision.

An attempt binds a versioned question snapshot, policy, time limit, randomization seed and accommodations. The server scores objective questions; manual review owns declared subjective items. Certification eligibility, issue, expiry, renewal, revocation and public verification are independent state machines.

7. Optional HF-managed AI

A future private-HF recommender may rank already-approved content from minimized, consented learner history. It is advisory; the learner can ignore it; deterministic catalog/eligibility policy filters output. Generative content must remain draft with human review and source attribution.

Permitted	Forbidden
content ranking, draft suggestions, optional explanation	scoring, certification, access, disciplinary action
private HF repo + endpoint + receipt	local inference or direct browser token
fairness/privacy monitoring	training on learner data without approval

8. Database and API contract

Target additions include content/version/attempt/credential/audit contracts and optional recommendation receipt references. Exact SQL waits for live schema diff. Current learning routes remain compatible while versioned endpoints are introduced additively.

- Tenant ownership and RLS on every learner/admin row.
- Optimistic concurrency or equivalent for mutable projections.
- Outbox/event records for notifications and integrations.

9. Security, privacy and accessibility

Apply least privilege, role separation, service-role-only state transitions, encrypted sensitive fields, audit and retention. Minimize learner event data; support consent/notice/export/deletion constraints. Target WCAG 2.2 AA and test keyboard, screen reader, focus, captions, color and timing accommodations.

- No raw assessment answers in routine analytics logs.
- Public credential verification reveals minimum necessary fields.
- Optional AI uses a separate privacy profile and kill switch.

10. Scaling and observability

Do not invent service objectives; establish baselines under measured load.

Pageinate catalog/admin views, cache published content, queue reminders/exports, rate-limit attempts and measure stale projections. Observe enrollment funnels, progress consistency, attempt errors, scoring latency, credential issuance, accessibility defects and optional AI acceptance/fairness separately.

11. Migration, tests and backlog

Add immutable content/attempt contracts, then dual-read/write, reconcile and enforce. Test state machines, tenancy, concurrency, scoring, accessibility, backups and client flows. The optional risk-event bridge is P6, after evidence taxonomy and consent controls.

- VT2-040 consented risk-event bridge.
- VT2-041 optional private-HF advisory recommender.
- VT2-042 live-schema-reviewed migration.

12. Definition of done

- Learning works during complete HF outage.
- Published content and attempt snapshots are immutable/versioned.
- Scores and credentials have deterministic evidence and audit.
- Tenant/RLS/accessibility/privacy tests pass.
- Any optional AI is advisory, private-HF, receipt-bound and independently killable.

References and package evidence

Primary sources and immutable model snapshots are listed in Research/Sources/source_ledger.csv and Research/Sources/HuggingFace/. Repository claims are traceable through repository_inventory.csv, source_file_audit.csv, runtime_call_graph.md, current_architecture_map.md and code_claim_ledger.csv.

- Architecture_Source_of_Truth.yaml - canonical invariants, roles, schemas and rollout.
- Model_Registry_Proposal.json - immutable role/deployment proposal.
- HuggingFace_Deployment_Matrix.csv - point-in-time provider and endpoint decisions.
- Technical_Findings_Register.csv / Implementation_Backlog.csv - evidence-backed remediation.
- Database_Target_Contract.md / Database_Migration_Specification.md - logical contract without fabricated SQL.
- Competitor_Analysis.md / Competitor_Matrix.csv - primary-source market design loop.

Authoritative external references

Hugging Face Inference Providers | Hugging Face Inference Endpoints | C2PA Technical Specification. Access dates, exact page URLs, claim summaries and verification status are preserved in the source ledger.

Universal operational controls

Security, privacy, cost, capacity, calibration, observability, incident response and rollback are release gates for every role. No UNKNOWN is replaced with an estimate; unresolved values remain explicit until an evidence-producing test or owner decision closes them.

Final boundary

Hugging Face-managed learned inference is authoritative. No learned production model runs inside the VeriTrust Vercel application or a VeriTrust-owned worker/server; no production weights are required in the VeriTrust deployment repository.